

Firmware Corruption

Adversaries may overwrite or corrupt the flash memory contents of system BIOS or other firmware in devices attached to a system in order to render them inoperable or unable to boot, thus denying the availability to use the devices and/or the system.^[1] Firmware is software that is loaded and executed from non-volatile memory on hardware devices in order to initialize and manage device functionality. These devices may include the motherboard, hard drive, or video cards.

In general, adversaries may manipulate, overwrite, or corrupt firmware in order to deny the use of the system or devices. For example, corruption of firmware responsible for loading the operating system for network devices may render the network devices inoperable.^{[2][3]} Depending on the device, this attack may also result in [Data Destruction](#).

ID: T1495

Sub-techniques: No sub-techniques

① **Tactic:** [Impact](#)

① **Platforms:** Linux, Network Devices, Windows, macOS

① **Impact Type:** Availability

Version: 1.3

Created: 12 April 2019

Last Modified: 12 May 2026

[Version](#) [Permalink](#)

Procedure Examples

ID	Name	Description
C0063	2025 Poland Wiper Attacks	During the 2025 Poland Wiper Attacks , adversaries performed a factory-reset on compromised devices that hampered forensic investigations. ^[4]
S0606	Bad Rabbit	Bad Rabbit has used an executable that installs a modified bootloader to prevent normal boot-up. ^[5]
S0266	TrickBot	TrickBot module "Trickboot" can write or erase the UEFI/BIOS firmware of a compromised device. ^[6]

Mitigations

ID	Mitigation	Description
M1046	Boot Integrity	Check the integrity of the existing BIOS and device firmware to determine if it is vulnerable to modification.
M1026	Privileged Account Management	Prevent adversary access to privileged accounts or access necessary to replace system firmware.
M1051	Update Software	Patch the BIOS and other firmware as necessary to prevent successful use of known vulnerabilities.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0167	Firmware Modification via Flash Tool or Corrupted Firmware Upload	AN0474	Firmware flash utility invoked with elevated privileges followed by raw access to firmware device path or changes to boot configuration.
		AN0475	Direct write access to /dev/mem or /sys/firmware combined with usage of firmware flashing utilities (e.g., flashrom).
		AN0476	EFI updates executed via system processes or binaries outside of expected patch windows or using unsigned firmware packages.
		AN0477	Firmware image uploaded via TFTP/SCP or web interface followed by reboot or unexpected loss of connectivity.

References

1. [Yamamura, M. \(2002, April 25\). W95.CIH. Retrieved April 12, 2019.](#)
2. [U.S. Department of Homeland Security. \(2016, August 30\). The Increasing Threat to Network Infrastructure Devices and Recommended Mitigations. Retrieved July 29, 2022.](#)
3. [CISA. \(2022, April 28\). Alert \(AA22-057A\) Update: Destructive Malware Targeting Organizations in Ukraine. Retrieved July 29, 2022.](#)
4. [CERT Polska. \(2026, January 30\). Energy Sector Incident Report – 29 December. Retrieved April 22, 2026.](#)
5. [Mamedov, O. Sinitsyn, F. Ivanov, A.. \(2017, October 24\). Bad Rabbit ransomware. Retrieved January 28, 2021.](#)
6. [Eclipsium, Advanced Intelligence. \(2020, December 1\). TRICKBOT NOW OFFERS 'TRICKBOOT': PERSIST, BRICK, PROFIT. Retrieved March 15, 2021.](#)